

Max Teeple

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Controls engineer for autonomous agricultural vehicles. The work runs from model-based control and vehicle dynamics through sensor fusion, inertial navigation, and object tracking, and it ends as embedded software on the machine.

PROFESSIONAL EXPERIENCE

Controls Engineer II

May 2023 – Present

CNH Industrial · Scottsdale, AZ

- Lead control development for autonomous and semi-autonomous agricultural vehicles, from a simulated strategy to software running on the machine.
- Design, simulate, and tune model-based controllers so the vehicle tracks its path and stays composed as speed, load, and the plan change.
- Close guidance on a fused navigation estimate built from inertial sensors and the rest of the navigation stack.
- Bring object tracks into the control problem, so what the vehicle perceives can change how it moves.
- Integrate that embedded stack into production platforms with the teams who own the machine, the sensors, and the perception software.

Research Engineer

May 2022 – May 2023

Raven Industries · Scottsdale, AZ

- Developed and validated embedded control software for real-time operation on precision-agriculture systems.
- Modeled vehicle guidance and stability, including the inertial-navigation and sensor-fusion path the controller depends on.
- Supported autonomy research on vehicle dynamics: how the platform should respond as the mission and the scene change.

Graduate Research Assistant

Fall 2020 – Spring 2022

Iowa State University · Ames, IA

- Developed model-based control for dynamic vehicle suspensions in partnership with John Deere.
- Carried the work through control design, simulation, and hardware-in-the-loop validation.
- Wrote the technical documentation and contributed to peer-reviewed research.

Product Engineering Intern

Summer 2021

John Deere · Waterloo, IA

- Built predictive machine-learning models to analyze engine emission signals.

Product Engineering Intern

Summer 2020

John Deere · Des Moines, IA

- Developed MATLAB and Simulink models for prototype machine-health systems.

Product Engineering Intern

Summer 2019

John Deere · Des Moines, IA

- Modeled electromechanical nozzle control systems in Simulink.

EDUCATION

M.S., Mechanical Engineering

Fall 2020 – Spring 2022

Iowa State University · Ames, IA

B.A., Physics and Computer Science

Fall 2016 – Spring 2020

Coe College · Cedar Rapids, IA

Minor in Mathematics

TECHNICAL SKILLS

Programming Python, Java, MATLAB, Simulink, C, C++

Tools Creo, SOLIDWORKS

Focus Autonomy, model-based design, control systems, vehicle dynamics, inertial navigation, sensor fusion, and object tracking